

Setup Guide OpenVPN Master branch

Version 0.5

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Introduction

On master branch *luci-app-openvpn* is replaced by *luci-proto-openvpn*.

There is no longer a separate entry in the GUI for OpenVPN but it has become a protocol so it is integrated in the interface.

The whole interface and inner working are redesigned so this guide is only applicable for **Master branch** as 25.12 and earlier branches use the old style *luci-app-openvpn*

This guide and also the OpenVPN implementation is very much a Work in Progress and subject to change.

Installation

If you do not have OpenVPN installed at all then install by first by updating APK and then installing *openvpn-openssl* and *luci-proto-openvpn*

Prerequisites

A fully functioning router with the correct time.

Prerequisites for an OpenVPN Server

Skip this paragraph for setting up an **OpenVPN Client** and **continue** to [Using Config File](#)

Check all involved subnets

For setting up an OpenVPN **tun** server **all three involved subnets have to be different and non overlapping**. So the routers subnet, the OpenVPN subnet and the Clients subnet all have to be different.

As you often cannot choose the subnet of the OpenVPN client, it is best to **avoid** using frequently used subnets for your router e.g. 192.168.1.1/24 or 192.168.0.1/24 but if you cannot easily change the routers subnet then leave it and hope for the best.

For the OpenVPN subnet 172.23.23.1/24 is chosen in this example which is not often used.

Check for Public IP address

To be able to connect from outside, your router must have a public IP address (either IPv4 and/or IPv6). Check if your router has a proper Public IPv4 address with (from command line):

`ifstatus wan | grep address`

A public IP address does **not** start with 192.168.X.X, 10.X.X.X, 172.16-31.X.X or a [CGNAT address](#) (IP addresses from 100.64.0.0 to 100.127.255.255)

Another way to test is to compare the routers IP address (`ifstatus wan | grep address`) with the address from [ipleak.net](#) it should show the same address for a Public IPv4 address. If it is not the same you might not have a Public IPv4 address and the router is not reachable with IPv4.

If your router is behind another router then:

- Check if that router has a Public IP address.
- Check if and how you can Port Forward from that router to your router which is going to run the OpenVPN Server.

If you are behind CGNAT, so do not have a public IPv4 address and do not have a public IPv6 address (check with: `ifstatus wan6`) or using IPv6 is not applicable then you have to involve a commercial third party to get a public IP address.

This can be a VPN provider which supports port forwarding (e.g. AirVPN, ProtonVPN), or you can rent a Virtual Private Server (I have an Oracle VPS which can be had for free, see at the bottom of this guide), or use things like [Netbird](#), [Zerotier](#), [Cloudflared](#), [Tailscale](#), [ngrok](#), [pinggy](#), or [tunnelmole](#) and there are more, I have setup Netbird on several OpenWRT and Windows and Linux clients and it works well, see my notes [about setting up Netbird on OpenWRT](#) and the [Netbird support thread](#).

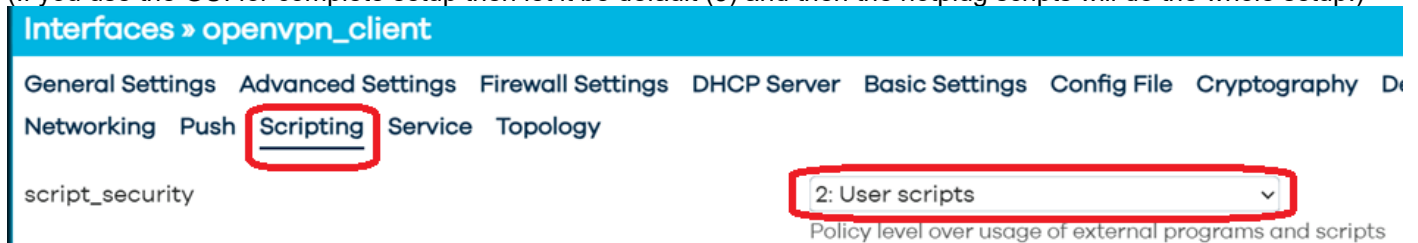
If your Public IP address is non static, e.g. it can change, then look into using [DDNS](#).

Test from outside

Proper **testing** can only be done **from outside** e.g. with your phone or laptop on cellular data or from a friends/neighbors internet.

Using Config File

When using your own config file to setup an **OpenVPN Client** set **script-security 2** (If you use the GUI for complete setup then let it be default (3) and then the hotplug scripts will do the whole setup.)



This will allow you to setup the VPN tunnel to you own liking

```
/var/run/openvpn.openvpn_client.conf:
cd /etc/openvpn/openvpn_client
status /var/run/openvpn.openvpn_client.status
```

```
syslog openvpn_openvpn_client
tmp-dir /tmp
script-security 2
dev tun1
dev-type tun
mtu-disc no
config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
```

If you leave the script security at its default which is 3 then the hotplug script will setup everything up for you. This reads the variables from the OpenVPN environment so it will setup a tunnel with a config file but no routing is setup as *route-noexec* is added to the config file.

For an **OpenVPN Server** you could try and leave the **script-security 3** as that will show traffic on the interface but you can not set routes via the script but for a simple setup this could work.

```
/var/run/openvpn.openvpn_client.conf with script-security 3:
cd /etc/openvpn/openvpn_client
status /var/run/openvpn.openvpn_client.status
syslog openvpn_openvpn_client
tmp-dir /tmp
script-security 3
dev tun1
dev-type tun
mtu-disc no
config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
setenv INTERFACE openvpn_client
script-security 3
setenv IPV6 '1'
ifconfig-noexec
route-noexec
up '/usr/libexec/openvpn-hotplug'
down '/usr/libexec/openvpn-hotplug'
route-up '/usr/libexec/openvpn-hotplug'
route-pre-down '/usr/libexec/openvpn-hotplug'
tls-crypt-v2-verify '/usr/libexec/openvpn-hotplug'
```

Setup Interface

To setup OpenVPN (whether client or server)

Network > Interfaces > Add new interface

Name: choose a name not more than 15 characters, I use *openvpn_client* in this example

Protocol: OpenVPN.

Add new interface...

Name

openvpn_client

Protocol

OpenVPN

Cancel

Create interface

Click:

Create interface

Now you can setup the OpenVPN interface

Interfaces » openvpn_client

General Settings

Advanced Settings

Firewall Settings

DHCP Server

Basic Settings

Config File

Cryptography

Devices

Keygen

Logging

Management

Networking

Push

Scripting

Service

Topology

Status

Device: openvpn-openvpn_client
Carrier: Absent
RX: 0 B (0 Pkts.)
TX: 0 B (0 Pkts.)

Protocol

OpenVPN

Disable this interface

☐

Bring up on boot

☒

Almost nothing here prevents you from selecting invalid configuration options which prevent openvpn from starting. Read the manual.
Options marked with ¹ are deprecated and will be removed.
Options marked with ² are OpenSSL only.

port

1194
TCP/UDP port # for both local and remote

Import your config file into the interface.

Click: Config File

Interfaces » openvpn_client

General Settings

Advanced Settings

Firewall Settings

DHCP Server

Basic Settings

Config File

Cryptography

Devices

Networking

Push

Scripting

Service

Topology

Drag en drop your OpenVPN config file in the box

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File Cryptography Devices Keygen Logging Management
Networking Push Scripting Service Topology

Clear

Clear

Drag and drop an ovpn config file here

Raw OVPN config

Dismiss

Save

Config files can contain all necessary info *inline* including keys certificates and username and password as shown below

Interfaces » openvpn_client

[General Settings](#) [Advanced Settings](#) [Firewall Settings](#) [DHCP Server](#) [Basic Settings](#) [Config File](#)
[Networking](#) [Push](#) [Scripting](#) [Service](#) [Topology](#)

Clear

Clear

Raw OVPN config

```
remote fr-rbx.vpnunlimitedapp.com
proto udp
port 1197

<auth-user-pass>
REDACTED
</auth-user-pass>

client
dev tun1
reneg-sec 0
persist-tun
ping 5
nobind
allow-compression no
remote-cert-tls server
auth-nocache
data-ciphers CHACHA20-POLY1305:AES-256-GCM:AES-128-GCM
auth sha512

pull-filter ignore "redirect-gateway def1"

#for hotplug scripts use 3:
script-security 3

|ca>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</ca>

<cert>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</cert>

<key>
-----BEGIN PRIVATE KEY-----
redacted
-----END PRIVATE KEY-----
</key>
```

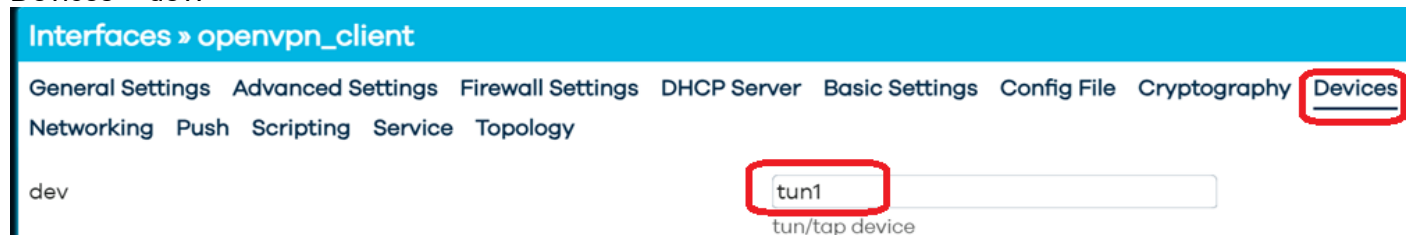
Often a config file just has *dev tun* so without a number, in this case OpenVPN itself will use the first available number which is **0** so the device will be tun0

I think this is bad practice and encourage you to choose a number for the device.

I use *tun1* for my client and *tun2* for my server.

As is chosen to setup the tunnel with a config file with script-security 2 there is no information about the used *l3_device* (the *tun1* device in this example) to make sure applications which use the interface can

extract the `I3_device` (e.g. `tun1`) from the config (e.g. for Policy Based Routing) also add the device under `Devices > dev`:



this way PBR wil now the device used by the openvpn_client interface

```
/etc/config/network/
config interface 'openvpn_client'
    option proto 'openvpn'
    option dev_type 'tun'
    option mtu_disc 'no'
    option multipath 'off'
    option config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
    option script_security '2'
    option dev 'tun1'
```

Setup Firewall

OpenVPN Client

Create a new firewall zone:

Network > Firewall > Add

Name: **vpn_client**

Input: **reject**

Output: **accept**

Intra zone forward: **reject**

IPv4 Masquerading: **Enabled** (ticked)

MSS clamping: **Enabled** (ticked)

Covered Networks: none

Allow forward from source zones: **lan**

Firewall - Zone Settings

General Settings Advanced Settings Contrack Settings

This section defines common properties of "vpn_client". The *input* and *output* options set the default policies for traffic. The *forward* option describes the policy for forwarded traffic between different networks within the zone. *Covered networks* defines the networks covered by this zone.

Name	vpn_client
Input	reject
Output	accept
Intra zone forward	reject
IPv4 Masquerading	☒ Enable network address and port translation IPv4 typically enabled on the <i>wan</i> zone.
MSS clamping	☒
Covered networks	unspecified

The options below control the forwarding policies between this zone (vpn_client) and other zones. *Destination zones* match forwarded traffic from other zones **targeted at vpn_client**. The forwarding rule is *not* imply a permission to forward from wan to lan as well.

Allow forward to <i>destination zones</i> :	unspecified
Allow forward from <i>source zones</i> :	lan

In this example I have adapted my config file to use: **dev tun1**

As we have used script-security 2 and use our own config there is no device information available for the firewall to use so we have to add the device on the Advanced Settings tab under Covered devices:

Firewall - Zone Settings

General Settings Advanced Settings Conntrack Settings

The options below control the forwarding policies between this zone (vpn_client) and other zones. *Destination zones* **vpn_client**. *Source zones* match forwarded traffic from other zones **targeted at vpn_client**. The forwarding rule is *unic* *not* imply a permission to forward from wan to lan as well.

Covered devices

 tun1 ▼

Use this option to classify zone traffic by raw, non-uci

For IPv6 also enable Masquerading IPv6 on the Advanced Settings Tab.

/etc/config/firewall:

```
config zone
    option name 'vpn_client'
    option input 'REJECT'
    option output 'ACCEPT'
    option forward 'REJECT'
    option masq '1'
    option mtu_fix '1'
    option masq6 '1'
    list device 'tun1'
```

```
config forwarding
    option src 'lan'
    option dest 'vpn_client'
```

Alternatively you can add an extra interface with name: *tun1_client*, protocol: *unmanaged* and device: *tun1*

You can use this interface *tun1_client* to add under Covered networks instead of adding the device under Covered devices

OpenVPN Server

Open up firewall for the server port

Create new zone *vpn_server*, Allow Input, Output and Intra zone forward

Allow from VPN to LAN , from VPN to WAN if you want internet for the connected clients and Allow from LAN to VPN if you want bidirectional traffic.

Status

View traffic with: `cat /var/run/openvpn.vpn_client.status` # replace *vpn_client* with the name of your interface

Management interface

```
#Add to openvpn config for management
management 127.0.0.1 14
management-log-cache 100
```

```
#for viewing: `nc 127.0.0.1 14`
```

Problems

When letting OpenWRT setup the routing via the hotplug scripts (script-security 3) there is no host route made

Furthermore it replaces the default route instead of using 128.0.0.0/1 and 0.0.0.0/1

OpenVPN can use a gateway although not per se necessary the hotplug routing does not take that into account

